

notation. If you aren't using a weight-specific data type that contains unit information, naming the variable something like `weight_kg` could be prudent. Indeed, in 1999 the Mars Climate Orbiter robotic space probe was lost when software supplied by Lockheed Martin produced calculations in imperial standard units, whereas NASA's systems used metric, resulting in an incorrect trajectory. The spacecraft reportedly cost \$125 million.

Sequential Numeric Suffixes in Names

Sequential numeric suffixes in your names indicate that you might need to change the variable's data type or add different details to the name. Numbers alone often don't provide enough information to distinguish these names.

Variable names like `payment1`, `payment2`, and `payment3` don't tell the person reading the code what the difference is between these payment values. The programmer should probably refactor these three variables into a single list or tuple variable named `payments` that contains three values.

Functions with calls like `makePayment1(amount)`, `makePayment2(amount)`, and so on should probably be refactored into a single function that accepts an integer argument: `makePayment(1, amount)`, `makePayment(2, amount)`, and so on. If these functions have different behaviors that justify separate functions, the meaning behind the numbers should be stated in the name:

`makeLowPriorityPayment(amount)` and `makeHighPriorityPayment(amount)`, OR
`make1stQuarterPayment(amount)` and `make2ndQuarterPayment(amount)`, for example.

If you have a valid reason for choosing names with sequential numeric suffixes, it's fine to use them. But if you're using these names because it's an easy choice to make, consider revising them.

Make Names Searchable

For all but the smallest programs, you'll probably need to use your editor or IDE's CTRL-F "find" feature to locate where your variables and functions are referenced. If you choose a short, generic variable name, such as `num` or `a`, you'll end up with several false matches. To make the name easy to find immediately, form unique names by using longer variable names that contain specific details.